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ORIGINAL PAPER

Cycles of *Euphausia superba* recruitment evident in the diet of Pygoscelid penguins and net trawls in the South Shetland Islands, Antarctica

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Abstract Size and sex of Antarctic krill taken from chinstrap and gentoo penguin diet were compared to those from scientific net surveys in the South Shetland Islands from 1998 to 2006 in order to evaluate penguin diet as a sampling mechanism and to look at trends in krill populations. Both penguin diet and net samples revealed a 4–5 year cycle in krill recruitment with one or two strong cohorts sustaining the population during each cycle. Penguin diet samples contained adult krill of similar lengths to those caught in nets; however, penguins rarely took juvenile krill. Penguin diet samples contained proportionately more females when the krill population was dominated by large adults at the end of the cycles; net samples showed greater proportions of males in these years. These patterns are comparable to those reported elsewhere in the region and are likely driven by the availability of different sizes and sexes of krill in relation to the colony.

Keywords Chinstrap penguin · Gentoo penguin · *Euphausia superba* · Recruitment · Sex-ratios · Spatial distribution

Introduction

Antarctic krill, *Euphausia superba*, are a keystone of the Antarctic ecosystem serving a critical role in the transfer of energy from macro-zooplankton to upper trophic level predators (Laws 1985). Krill are also the target of a major commercial fishery concentrated north of the South Shetland Islands (Everson and Goss 1991). Understanding the population dynamics of *E. superba* is critical to management of this ecologically and commercially important species. This topic is of particular interest since krill are ice-dependent and their populations have declined in recent decades of decreased sea-ice extent in the Antarctic Peninsula (Siegel and Loeb 1995; Loeb et al. 1997; Atkinson et al. 2004). In this study we look at krill demographic trends over 9 years in the South Shetland Island region based on the diet of two krill-dependent predators: chinstrap and gentoo penguins (*Pygoscelis antarctica* and *P. papua*).

Antarctic krill are relatively long-lived, living 6–7 years (Siegel 2000), and they generally begin to reproduce in their third or fourth summer (Siegel and Loeb 1994). Spawning occurs during the austral summer and krill may spawn multiple times during one summer (Ross and Quetin 1983). Algal growth on the underside of sea-ice is an important food source for adults during the early spring when accumulating reserves for reproduction (Quetin and Ross 2001). Furthermore, juvenile Antarctic krill are highly dependent on sea-ice algae during winter (Ross and Quetin 1991). Due to these relationships, krill populations are believed to fluctuate in relation to the extent and duration of winter sea-ice and the timing of sea-ice retreat (Smetacek et al. 1990; Siegel and Loeb 1995). Recently, as years of extensive winter sea-ice have become less frequent (Smith and Stammerjohn 2001), krill abundance appears to have declined due to limited recruitment of strong cohorts into the population (Quetin and Ross 2003).

To monitor fluctuations in the size-structure and abundance of krill populations, ship-based sampling has long been the standard method. While ship-based surveys are direct, they are costly and may be temporally or spatially

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limited. An additional measure of the status of krill populations is available from the diets of krill dependent predators (Croxall et al. 1988; Reid and Croxall 2001). Predator diet samples have different limitations than ship-based sampling and therefore provide a useful complement to ship studies. Pygoscelid penguins are abundant predators of *Euphausia superba* in the South Shetland Islands; they feed almost exclusively on krill during the chick-rearing period (Volkman et al. 1980; Jansen et al. 1998). Samples of their diet are relatively easily collected. Penguins are central place foragers during the breeding season; while feeding chicks they must return daily to the colonies and therefore must forage close to their breeding site. In order to understand the potential of penguins as indicators of krill population dynamics, given their restricted range, it is necessary to know to what extent the krill in their diet reflects the population of *E. superba* over a broader region.

Evaluation of penguins as samplers of krill by direct comparison to scientific ship-based sampling has been limited. On South Georgia penguins were found to take larger krill and more females than those captured in nets (Hill et al. 1996; Reid et al. 1996). The same was also true of krill eaten by fur seals *Arctocephalus gazella* in the South Shetland Islands (Reid et al. 2004; Osman et al. 2004).

In this study, we examine krill population dynamics in the South Shetland Islands based on the diet of chinstrap and gentoo penguins breeding at Cape Shirreff, Livingston Island. Krill dependent predators are monitored here as part of the US Antarctic Marine Living Resource program (AMLR), which also conducts ship-based studies of krill in the South Shetland Island region. Over 9 years we examined the size, age, and sex distribution of krill in penguin diets collected during the chick-rearing period of the austral summer and compared the results to those of the AMLR net-trawl surveys completed during the same times. We use this information to address the following questions: (1) how does the krill taken by penguins reflect the total population, as assessed by net trawls, and (2) accounting for the similarities and differences, what can we learn about changes in the population dynamics of *Euphausia superba*?

Materials and methods

Studies were conducted at Cape Shirreff, Livingston Island (62° 28′ S, 60° 47′ W; Fig. 1). Livingston Island is the second largest of the South Shetland Islands, located north of the tip of the Antarctic Peninsula. Cape Shirreff faces north into the Drake Passage, onto a broad continental shelf with the shelf break located 25–30 km offshore. Long-term monitoring of krill-dependent predators is conducted here by the US AMLR program in contribution to the Convention for the Conservation of Antarctic Marine Living Resources (CCAMLR). Chinstrap penguins feed chicks almost exclusively krill, >99% by mass. Gentoo penguins have a higher proportion of fish in the chick diet, but provisions to chicks still contain over 70% krill by weight (AMLR, unpublished data). Both species feed in open-waters within 5–30 km of the colony during the chick-rearing period.

Penguin diet samples were collected from 1998 to 2006 following CCAMLR standard protocols (Standard Methods 8a, b & c; CCAMLR 2004). Samples were taken from adult chinstrap and gentoo penguins returning from foraging trips to feed their chicks. Adult penguins returning from sea

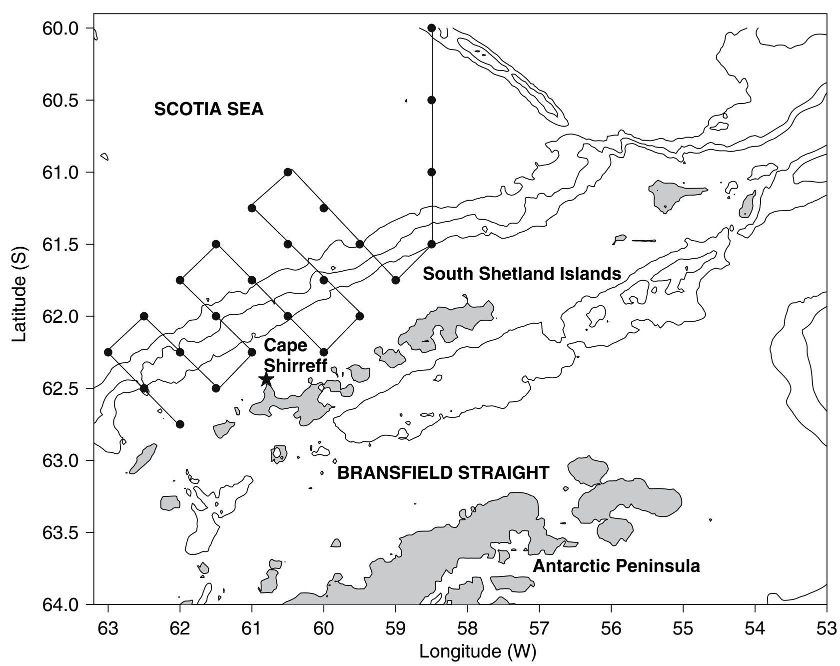

Fig. 1 Map of the South Shetland Islands, Antarctica and AMLR krill sampling stations. Cape Shirreff, Livingston Island, indicated by a *star*. Bathymetric contours shown are 1,000, 2,000, and 3,000 m

were followed back to the nest to confirm they were breeders; they were then captured, and their stomach contents were collected using the wet-offloading technique (Wilson 1984). Stomach contents were collected from five different adults on each sampling date. Chinstrap penguins were sampled once every 5 days from early January to mid-February for a total of 40 samples per year. Gentoo penguins were sampled once a week from mid-January to early February for a total of 20 samples per year. Sampling was conducted entirely during the chick-rearing period and was timed to coincide with the AMLR ship-based krill survey.

A sub-sample of 50 krill was removed from every diet sample containing intact krill. Occasionally fewer than 50 intact krill were found, and in these cases all intact krill were removed. Total length was measured to the nearest mm for each krill, from the anterior side of the eyeball to the tip of the telson. Krill 35 mm in length or smaller were considered juveniles based on CCAMLR standard protocols. Krill greater than 35 mm were visually sexed based on the presence or absence of a thelycom. Specific maturity stage was not determined.

Scientific net samples of krill were taken in mid-January to early February during each year of our study (for locations, see Fig. 1). These stations covered a larger and more offshore area than was typically used by penguins foraging during the chick-rearing period. Samples were collected using either an Isaacs-Kidd Midwater Trawl (IKMT) or Rectangular Midwater Trawl (RMT 8) towed obliquely from the surface to a maximum depth of 200 m. For further details on trawling methods see Siegel et al. (2002).

In net samples containing fewer than 100 krill, all individuals were measured. In larger samples 100–200 were measured. Krill were classified for sex and maturity stage based on Makarov and Denys (1981). This method does not assume krill smaller than 36 mm to be juveniles as was done in the penguin diet analysis; however, to make the net samples comparable to the penguin diet samples we considered all krill ≤35 mm as juveniles. This simplification creates an over-estimate of the proportion of juveniles: in the net trawls 44% of krill ≤35 mm were identified as adult stages and were therefore sexed. Categorizing krill ≤35 mm as juveniles could therefore bias the ratio of adult male to female krill if one sex was over represented in the smaller size classes. However, we found that the ratio of male to female krill in the net samples that were analyzed following the Makarov and Denys method was highly correlated with the male to female ration in these same samples when all krill ≤35 mm were considered juveniles ($F_{1,7} = 1686$, $R^2 = 0.99$, $P < 0.001$). Thus, the difference in methods did not alter the ratio of male and female krill.

For both species of penguin and for net trawls, we looked at the length-frequency distribution of krill in each year. Comparisons were made for all krill and for adult krill (>35 mm) only. Not all distributions were normal, so mean krill lengths were compared using non-parametric rank-sum tests. We also compared the ratio of males, females and juveniles found in the diet and in nets in each year and tested the relationship between these proportions and the average length of krill. Because we found an apparent bias toward female krill in the penguin diet samples in some years (see Results), we tested whether a higher proportion of females were in advanced maturity stages (based on shipboard data; maturity stages were not available for penguin diet data), and therefore more nutritionally valuable, in these years. All tests were two-sided and the level of significance was set at $P = 0.05$.

Results

The length of krill collected from all sampling methods (chinstrap penguins, gentoo penguins, and net trawls) varied inter-annually and showed a clear 4–5 year cycle of increasing krill size (Fig. 2). Krill taken during 1998 were relatively small, with a modal length of 36–40 mm. In each of the following 3 years, there was a progression in the modal size of krill taken, peaking in 2001 in the 51–55 m range. This same progression of modal sizes was repeated from 2003 to 2006, although the variance within and among sampling techniques was greater during this period. In 2002, krill lengths were quite variable and did not fit into the cycles seen in the other years; however, this year was in the transition between the two cycles described. In most cases the length-frequency distribution of krill taken was unimodal, though bimodal distributions were seen in several years for net samples, with a small peak in the juvenile (≤35 mm) range and a larger peak in the adult range.

Penguin diet samples and net samples had similar inter-annual trends in mean krill lengths (Table 1); nonetheless, differences in the krill sizes taken by these techniques existed. Krill taken by gentoo penguins were significantly larger than those taken by chinstraps in all years and significantly larger than those captured in nets in all years except 2000 and 2005 (Table 2). Differences in mean krill length from chinstrap penguin diet and net samples were significant in all years except 1999 and 2006; however, the direction of these differences was not consistent (Table 2).

Much of the difference in mean krill sizes between penguins and nets is accounted for by the absence of small krill in the penguin diet; krill smaller than 30 mm were rarely seen in the penguin diet (Fig. 2). When we examined the average size of adult krill (those >35 mm) found in penguin diet and in nets, the correlation was much stronger. Gentoo penguins still took krill 1–3 mm longer than those found in nets in most years, but chinstrap penguins and nets

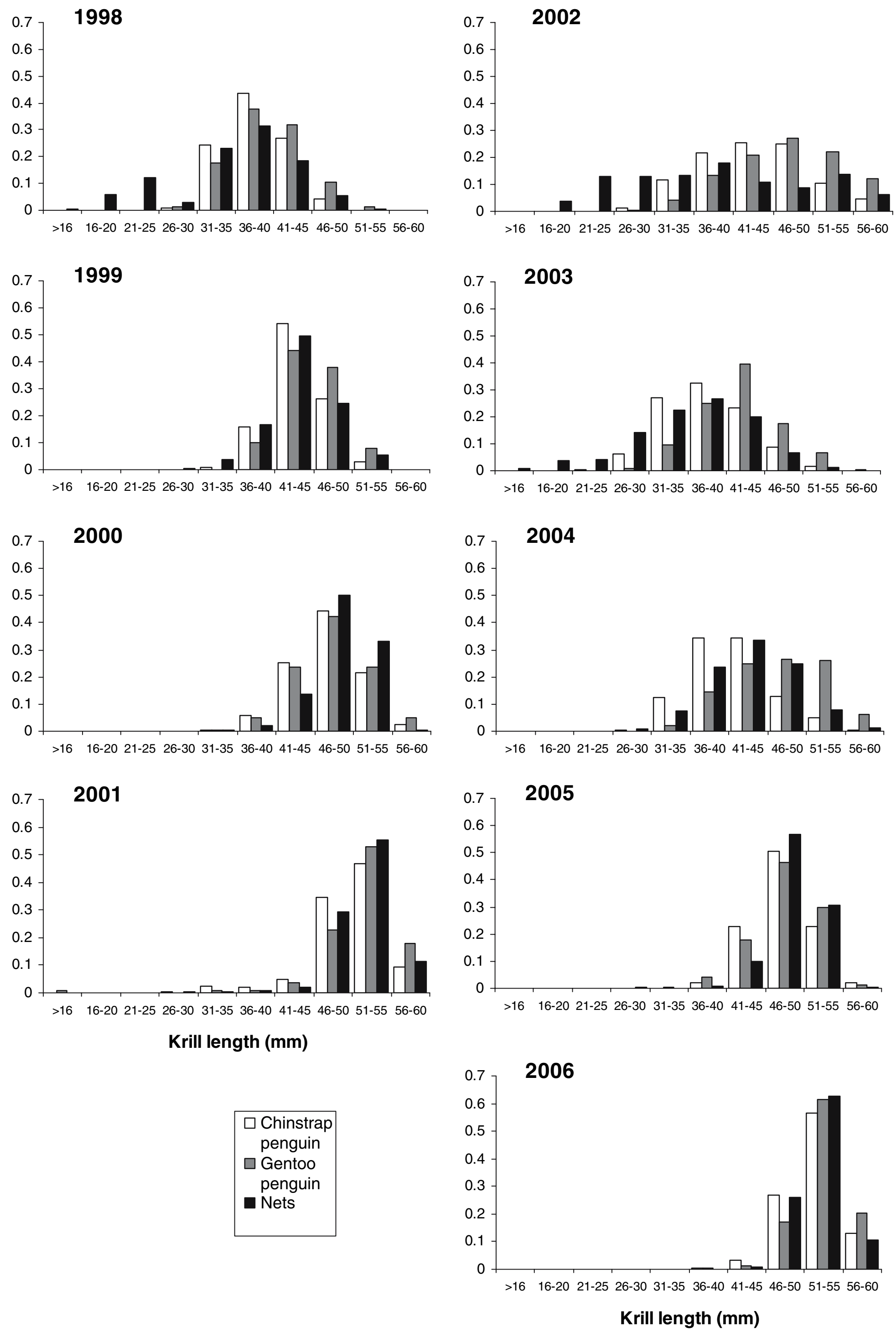

Fig. 2 Length-frequency distribution of Antarctic krill *Euphausia superba* from net samples in the South Shetland Islands and from diet samples of chinstrap and gentoo penguins *Pygoscelis antarctica* and *P. papua* at Cape Shirreff, Livingston Island, 1998–2006

contained krill of approximately the same mean lengths (Table 1). Significant statistical differences were detected in 4 years (2000, 2001, 2004, 2005), with chinstrap penguins taking smaller adult krill than those captured in nets; however, the differences in means (0.8, 1.1, 1.1, and 1.6 mm respectively) were within or only slightly greater than our measurement error (Table 2).

The proportion of adult females, adult males, and juvenile krill found in both penguins diet and in the net trawls varied among years. However, the pattern of variation was quite different in the penguin diets than the net trawls (Fig. 3). In the chinstrap penguin diet, the ratio of females to males increased over the first 4 years of study, as the average length of the krill increased. In 2002 the ratio of

Table 1 Mean lengths of Antarctic krill *Euphausia superba* (mm; mean ± SD) from diet samples of chinstrap and gentoo penguins *Pygoscelis antarctica* and *P. papua* and from net trawls in the South Shetland Islands, 1998–2006

		Mean krill length (mm) ± SD		
		Chinstrap	Gentoo	Nets
All krill	1998	38.6 ± 4.0 *n* = 1954	40.1 ± 4.5 *n* = 517	35.1 ± 7.8 n = 2364
	1999	43.7 ± 3.5 *n* = 1936	45.4 ± 3.7 *n* = 518	43.5 ± 4.3 *n* = 433
	2000	47.6 ± 4.4 *n* = 1914	48.1 ± 4.6 *n* = 703	48.6 ± 3.7 *n* = 182
	2001	50.5 ± 4.8 *n* = 1916	51.8 ± 5.6 *n* = 626	51.6 ± 3.8 *n* = 922
	2002	43.5 ± 6.8 *n* = 2000	47.4 ± 6.7 *n* = 843	38.1 ± 11.2 *n* = 532
	2003	38.3 ± 5.4 *n* = 1774	42.2 ± 5.4 *n* = 532	35.8 ± 7.6 *n* = 1279
	2004	41.4 ± 5.1 *n* = 1765	47.0 ± 5.9 *n* = 893	43.1 ± 5.6 *n* = 511
	2005	48.0 ± 3.6 *n* = 1920	48.4 ± 4.1 *n* = 644	48.9 ± 3.7 *n* = 510
	2006	52.1 ± 3.3 *n* = 1882	53.1 ± 3.1 *n* = 957	52.1 ± 2.7 *n* = 530
Adult krill	1998	40.3 ± 3.1 *n* = 1462	41.6 ± 3.5 *n* = 339	40.5 ± 3.5 *n* = 1317
	1999	43.8 ± 3.4 *n* = 1916	45.4 ± 3.7 *n* = 517	44.0 ± 3.8 *n* = 415
	2000	47.6 ± 4.3 *n* = 1906	48.1 ± 4.5 *n* = 699	48.7 ± 3.5 *n* = 181
	2001	51.0 ± 3.9 *n* = 1865	52.3 ± 3.8 *n* = 617	51.8 ± 3.3 *n* = 916
	2002	45.1 ± 5.7 *n* = 1735	48.0 ± 6.1 *n* = 805	46.2 ± 7.2 *n* = 305
	2003	41.3 ± 4.0 *n* = 1179	43.3 ± 4.5 *n* = 475	41.4 ± 3.9 *n* = 699
	2004	42.5 ± 4.5 *n* = 1544	47.3 ± 5.6 *n* = 873	44.1 ± 4.8 *n* = 467
	2005	48.0 ± 3.6 *n* = 1919	48.4 ± 4.0 *n* = 641	49.1 ± 3.0 *n* = 505
	2006	52.1 ± 3.3 *n* = 1882	53.1 ± 3.1 *n* = 957	52.1 ± 2.7 *n* = 530

females to males decreased sharply, then again increased over the following 4 years. The inter-annual pattern in gentoo penguin diets was virtually the same as in chinstrap diet samples, although the proportion of female krill was always higher. For both species, a significant positive relationship existed between the average length of krill and the proportion of female krill in the diet (chinstraps: $F_{1,7} = 19.34$, $R^2 = 0.73$, $P = 0.003$; gentoos: $F_{1,7} = 7.33$, $R^2 = 0.51$, $P = 0.03$). In contrast, the ratio of female to male krill in net trawls remained roughly equal or contained a slightly higher proportion of males throughout the study period; nets showed no relationship between the average krill length and the proportion of females ($F_{1,7} = 0.077$, $R^2 = 0.10$, $P = 0.40$). The higher proportion of female krill in chinstrap and gentoo penguin diet samples was not related to the proportion of females in advanced maturity stages (% advanced maturity stages from AMLR 2006; chinstraps: $F_{1,6} = 0.87$, $R^2 = 0.13$, $P = 0.39$; gentoos: $F_{1,6} = 0.38$, $R^2 = 0.06$, $P = 0.88$).

In years when juvenile krill appeared in the net trawls, juveniles also appeared in the penguin diet (Fig. 3). However, the proportion of juveniles was generally greater in the net trawls than in the chinstrap or gentoo penguin diet.

Discussion

Penguin diet and net samples showed similar inter-annual trends in the length of Antarctic krill: a progression of modal krill length over 4–5 year cycles was evident in both cases. Length is a useful proxy for age; therefore, we can interpret the same pattern of recruitment and cohort strength from both methods. Penguin diet and net samples differed, however, in the ratio of male to female krill sampled. Penguin diet samples were dominated by female krill when krill were longer while net samples often contained more male krill in these years. The difference in sex ratios may reflect biases of the two sampling methods. Alternately, since the foraging area of penguins is generally inshore of the net sampling stations, the different sex ratios may indicate different spatial distributions of male and female krill.

Comparison of penguin diet and net samples:

Chinstrap penguins proved to be good indicators of the size of Antarctic krill available, as compared to the net trawls, excepting the first year of recruitment for a new cohort (e.g., 1998 and 2002). In these years the length distributions differed due to the absence of small krill (<30 mm) in the penguin diet; nonetheless, the average adult krill taken by chinstraps were consistently similar in size to the average adult caught in nets. Statistical differences could be detected in some cases; however, the differences in means generally fell within the measurement error (±1 mm) of the two sampling methods suggesting that the differences are not biologically meaningful. Gentoo penguins consistently took larger krill than chinstrap penguins or the net trawls, averaging 1–3 mm longer. In all cases, the differences among years were notably larger than the differences among sampling methods within a year. Furthermore, with each sampling method, the same progression of modal sizes over 4–5 year cycles is evident.

Small krill were generally absent from the penguin diet even when present in substantial numbers in the net trawls.

Table 2 *Z*-values of rank-sum tests comparing mean lengths of Antarctic krill *Euphausia superba* from diet samples of chinstrap and gentoo penguin *Pygoscelis antarctica* and *P. papua* and from net trawls in the South Shetland Islands, 1998–2006

	All krill			Adult krill		
	Chinstrap vs. gentoo	Chinstrap vs. nets	Gentoo vs. nets	Chinstrap vs. gentoo	Chinstrap vs. nets	Gentoo vs. nets
1998	−5.97*	−13.93*	−12.35*	−5.78*	**0.46**	−5.06*
1999	−8.96*	**−0.15**	−6.45*	−8.76*	**0.90**	−5.57*
2000	−2.43*	3.54*	**1.78**	−2.50*	3.60*	**1.79**
2001	−7.80*	5.93*	−3.02*	−7.72*	5.21*	−3.33*
2002	−13.20*	−10.62*	−15.04*	−11.06*	**1.94**	−4.01*
2003	−13.63*	−8.20*	−16.57*	−8.58*	**0.76**	−7.47*
2004	−22.03*	6.91*	−11.32*	−20.05*	6.72*	−10.08*
2005	−3.57*	6.30*	**1.83**	−3.74*	6.67*	2.00*
2006	−7.76*	**−0.54**	−6.74	−7.76*	**−0.54**	−6.74*

Significant differences ($P < 0.05$) indicated with *. Non-significant differences highlighted in bold

Other studies have found that marine birds and seals feed on larger krill than those caught in nets (Reid et al. 1996; Osman et al. 2004). Because these predators capture krill individually, it is likely that feeding on larger krill is more efficient. Since penguins are visual predators (Wilson et al. 1993), larger krill may also be more easily located. The chinstrap penguins on Livingston Island did not feed on small krill; however, they took adult krill that were either similarly sized or smaller than those caught in net samples. This evidence indicates that size selection is occurring against the smallest krill, but that no preference is made after krill reach a threshold size. Gentoo penguin diet samples, in contrast, contained slightly larger krill on average than chinstrap penguin or net trawl samples in all years. This size selection may reflect differences in foraging behavior: gentoo penguins generally dive deeper and forage within a closer range of the colony than chinstrap penguins; or, it may reflect the higher energetic demands of rearing gentoo chicks versus chinstrap chicks (Trivelpiece et al. 1986).

The other major difference between the penguin diet and net trawls was that the proportion of females found in the penguin diet increased in years with larger krill sizes while the net samples contained either equal numbers of males and females or more males in these years. A bias towards female krill was also found in comparisons between penguins and nets in the South Georgia region. Hill et al. (1996) suggested that macaroni penguins *Eudyptes chrysolophus* may take more adult female krill either because they are more nutritionally valuable or because adult male krill may be better at escaping from predators. Reid et al. (1996), when looking at a similar pattern with both macaroni and gentoo penguins, considered both of these hypotheses and also suggested that penguins may simply select for the largest krill, which are generally females. Indeed, we found that when krill were large, penguins exhibited a bias towards female krill. In years when krill were small and therefore not expected to differ in nutritional content or ability to escape predators, penguins did not exhibit the bias towards females.

While these factors may play a role in the selection of krill, they are likely secondary causes. The nutritional difference between male and female krill is greatest when females are gravid (Clarke 1980, 1984), but the years when penguins consumed the most female krill did not correlate with the years when most female krill were in advanced maturity stages. Alternatively, if krill length alone were the driving factor in krill selection, we would expect chinstrap penguins to take consistently larger adult krill than those caught in nets, but this was not the case (Fig. 2). We would also expect the mean size of krill taken by both penguin species to remain large and to have little interannual variation. Although that may not have been possible if larger krill were absent from the population in some years, we might nonetheless expect penguin-breeding performance to suffer as a result. However, this was not the case in this population (Hinke et al. 2007). Furthermore, krill are not the only possible prey item for penguins. Both species do capture some fish (Jansen et al. 1998, AMLR, unpublished data), and at other times of the year may depend on it heavily (Lescroel et al. 2004). Fish are likely to be faster swimmers than krill, but penguins are clearly capable of feeding on them.

We propose that the difference in sex ratios between penguin diet and net samples is more likely due to the spatial distribution of different sex and maturity stages of krill. Most of the ship survey is conducted further offshore than the penguins typically foraged during this time. Male and female krill are known to aggregate separately, as are different maturity stages of krill (Watkins et al. 1992). In general, females are thought to move offshore to spawn, and then move back inshore in the fall after spawning has ceased (Siegel 2000). While previous work found adult females further offshore in this region (Trathan et al. 1993), preliminary analysis for some years of our study has found more mature males offshore than females (AMLR 2003, 2004,

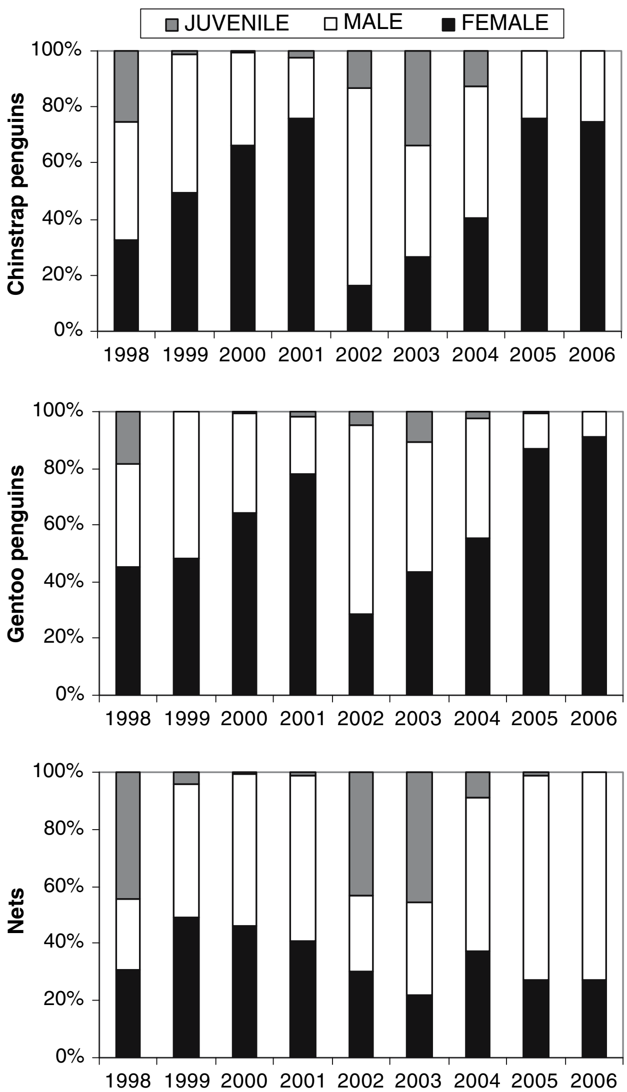

Fig. 3 Proportions of adult female, adult male, and juvenile Antarctic krill *Euphausia superba* in net samples from the South Shetland Islands and in diet samples from chinstrap and gentoo penguins *Pygoscelis antarctica* and *P. papua* at Cape Shirreff, Livingston Island, 1998–2006

2005). The abundance of females found in the penguin diets further raises the question of whether female krill were more likely to be found near-shore on the continental shelf, and if so does this indicate that fewer females are spawning, or are they are using the shelf region pre- or post-spawning?

Similarly, certain sex/maturity stages may have differing vertical distributions in the water column. Krill must balance the increased foraging benefit near the surface with the increased predation risk (De Robertis 2002). The energetic demands of reproduction might make females more likely to spend time at the surface despite predation risk; in a northern euphausiid species, *Meganyctiphanes norvegica*, female krill were more likely to be found close to the surface than males during the night (Tarling 2006). Net tows in this study were conducted down to 200 m, while the penguins typically foraged from the surface to 100 m (AMLR, unpublished data).

Furthermore, some sex/maturity stages may be found at more predictable locations than others based on the bathymetric or hydrographic habitat. Predictable features are likely to be targeted by penguins, while the ship survey instead follows a fixed grid. These fine scale spatial differences cannot be detected from our current study, and depend on further knowledge of the behavior of different sex/maturity classes of krill. Recent ongoing studies of near-shore krill distribution may help elucidate reasons for the sex ratio differences between penguin and net samples.

Krill population cycles:

Chinstrap and gentoo penguin diet samples and net samples all show the same clear pattern of 4–5 years of increasing modal krill lengths. Based on expected growth patterns for Antarctic krill (Siegel 1987), this progression indicates that a single large cohort, or two back-to-back strong cohorts, has been forming the bulk of the population. In both cycles we see a strong juvenile influx in the first year of the cycle, most evident in the net samples. By the second year the cohort strength is evident in penguin diet as well. The cohort(s) continue to move through the population until they are 5+ before the next large cohort recruits into the population and the older group disappears from the samples.

Episodic recruitment of large juvenile cohorts of Antarctic krill has been well established around the Scotia Sea and the Antarctic Peninsula. The leading hypothesis continues to be that the reproductive and recruitment success of krill is largely driven by the extent and duration of winter sea-ice (Siegel and Loeb 1995; Loeb et al. 1997). The annual reproductive output of krill can vary up to tenfold depending on how much reserves they accumulate in the spring, and algal growth on the bottom of sea ice is an important source of food during this time (Quetin and Ross 2001). Furthermore, young of the year krill depend heavily on ice-algae to survive their first winter (O'Brien 1987; Marschall 1988; Melnikov and Spiridonov 1996). A warming trend has been observed in the region over the last 50 years concomitant to decreased winter sea-ice (Fraser et al. 1992; King 1994; Smith and Stammerjohn 2001). These changes have been linked to declines in krill biomass in recent decades (Loeb et al. 1997; Atkinson et al. 2004).

Over the last decade, patterns of episodic recruitment have been demonstrated for krill populations around the Antarctic Peninsula. Fraser and Hoffman (2003) looked at samples of Adélie penguin *Pygoscelis adeliae* diet and found that one or two back-to-back cohorts of krill sustained the population over a 4–5 year cycle from 1988 to 1997. The end of their study period showed modal krill lengths of 31–35 mm, consistent with the mode of 36–40 mm seen in our study the following season in 1998. Quetin and Ross (2003) examined net samples of krill from 1992 to 2002 and found two cycles of two successive strong year classes followed by 3–4 years of poor recruitment. The later half of this study overlapped with our study period and the same modal peaks in krill length are seen in both locations. While there is debate over whether the krill found in these different regions are all from the same population (see Siegel et al. 2003), it is clear that the same environmental forces are affecting krill in many regions, and that the demographic trends are similar on a broad scale.

Episodic recruitment of large cohorts is uncommon among zooplankton, but quite common among fish populations. Due to the long life span of Antarctic krill, it is not surprising that they may exhibit similar demographic and lifehistory patterns as small pelagic fish. Infrequent new cohorts may be a normal and healthy pattern. However, if new strong cohorts are only being introduced at the end of the lifespan of the previous cohort, then the population is quite vulnerable should the time between successful cohorts be pushed apart any further.

As demonstrated in this study, penguin diet samples are an effective indicator of certain large-scale population trends of Antarctic krill. Penguin diets are limited in that recruitment events may not be detected in the first year; nonetheless, the relative strength of different cohorts is evident in subsequent years. The same pattern of episodic recruitment seen in net trawls is clear in the penguin diet. In contrast, penguin diet and net samples show differing trends in krill sex ratios when krill are largest. Penguin diet samples, which come from a relatively small near-shore region, contain far more females than the net samples, which come from a larger off-shore region. Since adult females are expected to move off-shore to spawn, this finding raises the question of whether females were not spawning in these years, or whether they were using the near-shore habitat pre- or post-spawning. The differences between penguin and net samples highlight the value of looking at predator diets in conjunction with net tows to evaluate krill population dynamics. Further analysis of the spatial distribution of size and sex/age classes of krill and the association between these patterns and sea-ice and oceanographic parameters may help our understanding of how penguin diet reflects the total population and how ocean climate variability drives krill populations.

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